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A common novel splice variant of *SLC22A1* (*OCT1*) is associated with impaired responses to imatinib in patients with chronic myeloid leukaemia

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Summary

Approximately one-third of patients with chronic myeloid leukaemia will fail to achieve or maintain responses to imatinib. Changes in solute carrier family 22 (organic cation transporter), member 1 (SLC22A1, also termed OCT1), the main transporter for imatinib, have been proposed as a possible predictive factor. We analysed *SLC22A1* mRNA levels and single nucleotide polymorphisms (SNPs) located in exon 7 in 153 diagnostic whole blood samples from two patient cohorts. The level of *SLC22A1* expression did not significantly correlate with imatinib failure or achievement of molecular remission. The SNP 408V>M (g.1222G>A) was present in 65% of patients and was associated in all cases with an eight base-pair insertion (8^+ allele) at the 3′ end of exon 7. The latter generates an alternative splice site, leading to a premature stop codon. M420del was found in 33% of patients and never in *cis* with 8^+ (the 3^- allele). Significantly longer times to 1% and 0·1% molecular responses (by quantitative reverse transcription polymerase chain reaction) were seen in patients with 8^+8^+ or 8^+N compared to those with the remaining four genotypes (N = no insertion or deletion). Patients lacking 8^+ and 3^- (NN, 18%) showed the best outcomes overall. Thus, while *SLC22A1* expression does not appear to affect response, alterations in its splicing or amino acid sequence may do so.

Keywords: chronic myeloid leukaemia, tyrosine kinases, prognostic factors, drug resistance.

Chronic myeloid leukaemia (CML) is characterized by a reciprocal translocation between chromosomes 9 and 22 [t(9;22)(q34;q11)], creating the Philadelphia chromosome and bringing together the Breakpoint Cluster Region (*BCR*) and Abelson (*ABL1*) genes. The protein encoded by this fusion gene is a dysregulated tyrosine kinase that phosphorylates multiple downstream proteins resulting in changes in cell proliferation, DNA repair, differentiation and cell cycling. The tyrosine kinase inhibitors (TKI), of which imatinib is the most widely used, block BCR-ABL1 tyrosine kinase activity by the competitive inhibition of ATP binding, resulting in a dramatic reduction in the differentiation and proliferation of *BCR-ABL1* positive cells.

The majority of patients treated with imatinib achieve haematological, cytogenetic and molecular remissions and are unlikely to progress to blastic transformation. However, a minority will fail to respond and a further subset of patients loses their initial response. The ability to identify these patients early would be advantageous, as it would help clinicians to optimize therapies on an individual basis. The mechanisms underlying imatinib resistance however, are not fully understood but include inadequate compliance, mutations in the *BCR-ABL1* tyrosine kinase domain and changes in intracellular pathways independent of BCR-ABL1 signalling (Apperley, 2007; Marin *et al*, 2010). A number of studies have investigated whether variations in the expression and function of solute carrier family 22 (organic cation transporter), member 1 (SLC22A1, also termed OCT1), which is thought to transport imatinib into the cell, determines responsiveness to imatinib. A number of studies have described an association between high *SLC22A1* expression (measured by real time quantitative polymerase chain reaction

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[qPCR]) and improved outcomes (Crossman *et al*, 2005; Wang *et al*, 2008; Dulucq & Krajinovic, 2010; Marin *et al*, 2010; Nardinelli *et al*, 2012; Zhong *et al*, 2012). This, however, has not been substantiated by other investigators (White *et al*, 2007; Zhang *et al*, 2009; Razga *et al*, 2011). Conflicting results have also been seen when correlating the presence of single nucleotide polymorphisms (SNPs) in *SLC22A1* with outcome (Zach *et al*, 2008; Kim *et al*, 2009; Bazeos *et al*, 2010; White *et al*, 2010a,b; Maffioli *et al*, 2011; Giannoudis *et al*, 2012). Some, but not all of these discrepancies might be explained by the use of different starting populations of cells (whole blood, mononuclear cells or a selected CD34+ population) and/or by different methods of measuring *SLC22A1* expression or activity.

The present study examines the relationship between *SLC22A1* expression/SNPs and imatinib response, in an attempt to resolve these conflicting findings.

Methods

Patient and sample selection

Patients were selected retrospectively from two cohorts – patients from the UK National Cancer Research Institute (NCRI) UK STI571 ProspectIve RandomIzed Trial (SPIRIT-1) treated exclusively with imatinib (with either 400 or 800 mg daily) and a cohort of imatinib-treated patients from the Imperial College Healthcare NHS Trust (ICHNT). All patients consented to the use of their diagnostic material. Inclusion criteria were as follows: a confirmed diagnosis of CML, peripheral blood taken for analysis prior to initiation of imatinib, no previous treatments (except hydroxycarbamide) and follow-up data available for at least 6 months.

Laboratory assessments

BCR-ABL1 transcripts were measured in whole blood for minimal residual disease monitoring using reverse transcription qPCR (RT-qPCR) using *ABL1* as a reference gene, as reported previously (Foroni *et al*, 2011; Gerrard *et al*, 2011).

SLC22A1 and *BCR-ABL1* transcript levels, as well as those of two control genes: beta2-microglobulin (*B2M*) and beta-glucuronidase (*GUSB*), were quantified using pre-treatment diagnostic cDNA from whole blood leucocytes by RT-qPCR using a Taqman system on a StepOnePlus (Applied Biosystems, Paisley, UK) in fast mode (45 cycles). cDNA samples had been stored at −80°C. Primers and probes were designed using ABI PRIMER EXPRESS 3 software with sequence data from Ensembl (http://www.ensembl.org/Homo_sapiens/Gene?g=ENSG00000175003) and *in silico* validation from the University of California, Santa Cruz (http://genome.ucsc.edu/). Only sequences in which there were no known SNPs were used. These were as follows: *SLC22A1* Forward (exon 6) GCA CCT TCA TCC TGA TGT ACC TG, *SLC22A1* Reverse (exon 7) CGG AGT AAA GGA AAT CCA GGT AGA, *SLC22A1* Probe 6-Carboxyfluorescein (6FAM)- CCT GCA CAT GGG CGC CAC C; *B2M* Forward GAG TAT GCC TGC CGT GTG, *B2M* Reverse ATC CAA ATG CGG CAT CT, *B2M* Probe VIC-TCC ATG ATG CTG CTT AC-MGB; *GUSB* Forward GAA AAT ATG TGG TTG GAG AGC TCA TT, *GUSB* Reverse CCG AGT GAA GAT CCC CTT TTT A, *GUSB* Probe VIC-ACT CTC GTC GGT GAC TG-MGB.

Absolute transcript levels were then derived using standard dilution curves for each gene and expressed as the ratio of the target gene to the geometric mean of the two reference genes (Vandesompele *et al*, 2002). This analysis was performed in duplicate for each sample and also on cDNA from K562 cell lines (as a positive control) and non-template controls (NTC).

Genomic DNA was extracted from stored GTC-lysate remission/pre-transplant specimens using Qiacube automated platform (Qiagen, Manchester, UK). PCR was carried out to amplify exon 7 of the *SLC22A1* gene using Qiagen Fast Cycling Taq master mix. Primers were as follows: forward GTC TCT GAC TCA TGC CTT TGA CTT G, reverse CAT CTT TGT TCT CAT TCC AGA GGC. Samples were then diluted, sequenced using Big Dye Terminator v3.1 cycle sequencing kit, purified using Big Dye X Terminator purification kit and analysed on an ABI 3500 platform. cDNA samples were processed similarly, using the following primers: forward (exon 7) CGG ACT CTG TGC TCT ATC AGG, reverse (exon 8) GTA CAG CTC AGC ATT CAC CA. In addition 6FAM-labelled reverse primers were used to enable fragment length analysis using high performance capillary electrophoresis (Genescanner, ABI 3130 apparatus; Applied Biosystems, Paisley, UK). Associations between SNPs and insertions/deletions were subsequently confirmed in a further cohort of 354 patients from the UK National Cancer Research Institute (NCRI) UK SPIRIT-2 trial.

Endpoints and statistical analyses

Using an externally derived conversion factor, *BCR-ABL1/ABL1* values were converted to the international scale (IS) to enable time to residual disease levels of 10%, 1% (taken as roughly equivalent to complete cytogenetic response (CCyR: Hughes *et al* (2010)) and 0·1% (major molecular response, MMR) values to be calculated (Cross, 2009). Imatinib failure-free survival was calculated by defining an event as: loss of haematological or cytogenetic response, progression to advanced phase or blast crisis, stem-cell transplant, death from any cause or change of imatinib to a second generation TKI due to suboptimal response (according to European Leukaemia Net criteria (Guilhot *et al*, 2012), or SPIRIT-1 trial protocol). The median expression level for each target gene/reference was used to divide patients into equal 'high' or 'low' expressing groups (as in Marin *et al* 2010, Nardinelli *et al*, 2012 and Wang *et al*, 2008). However, since there is no reason *a priori* to assume the most biologically significant threshold lies at the median, we developed an algorithm

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(implemented in the statistical programming environment, R) that calculates the logrank statistic for each possible threshold. This was applied to each cohort in a secondary analysis to determine whether the optimal threshold could be validated in the second cohort. The R code for this algorithm can be obtained from the corresponding author. Kaplan-Meier curves and the Mantel-Haenszel test/logrank test were used to compare the probabilities of reaching pre-defined endpoints over time for the two groups and Fisher's exact test was used to compare proportions. Multivariate analysis was performed using Cox proportional hazard modelling using the following variables: *SLC22A1* genotype, *SLC22A1* expression (high or low), sex, age, imatinib dose (400 or 800 mg), Sokal score and patient cohort. GRAPHPAD 4 PRISM 4.02 and R were used for all statistical analyses and a cut-off of 0·05 was used for significance testing.

Results

Patient characteristics

Forty-nine patients referred to Imperial College Healthcare NHS Trust (ICHNT cohort) fulfilled the inclusion criteria had sufficient diagnostic material for gene expression analysis, of which 47 had sufficient material for further SNP analysis. Of the SPIRIT-1 cohort, 103 were available for gene expression analysis and 109 had material available for SNP analysis. Demographic and clinical data for these patients are shown in Table I.

Validation of reference genes

The Normfinder algorithm (Andersen, 2004) was used to assess the stability of *GUSB* and *B2M*, alongside *ABL1* values derived at presentation in the SPIRIT-1 cohort. This algorithm takes into account both intra- and inter-group stability for each gene. *B2M* was found to have the best stability score (0·121 vs. 0·162 and 0·164), and the geometric mean of *B2M* and *GUSB* was found to be the optimally stable variable (0·110). *B2M* was also found to be the most stably expressed gene (0·183, compared to 0·332, 0·222 and 0·213) in a sample of 22 patients that compared *B2M*, *ABL1*, *GUSB* and ribosomal 18S (*RPS18*).

Table I. Demographic and clinical data of Imperial College Healthcare NHS Trust (ICHNT) and STI571 ProspectIve RandomIzed Trial (SPIRIT)-1 cohorts.

	ICHNT Group	SPIRIT-1 Group
Number	49	109
Median Age (range), years	44·3 (20–86)	55·2 (19–80)
Male: Female	26:23	68:41
Median follow-up (range), years	5·24 (1·4–10·24)	3·58 (0·72–6·25)
Achievement of RT-qPCR of <1% at 12 months on imatinib	65%	74%
Achievement of RT-qPCR of <0·1% at 12 months on imatinib	31%	39%
Number of imatinib failure events	21 (43%)	26 (24%)

RT-qPCR, real time quantiative reverse transcription polymerase chain reaction.

SLC22A1 *and neutrophil count*

Previous studies have highlighted the fact that *SLC22A1* is differently expressed in different cell types and therefore *SLC22A1* expression level by RT-qPCR may just be a surrogate marker for differing proportions of each cell type in patient samples (Bazeos *et al*, 2010; Engler *et al*, 2010; Racil *et al*, 2010). The neutrophil count at diagnosis was documented in 90 SPIRIT-1 patients. No correlation was found between neutrophil percentage and *SLC22A1* level by RT-qPCR in these patients (r^2 = 0·12, P = 0·3).

SLC22A1 *expression and response to imatinib*

Using the median expression level by RT-qPCR (0·4%; range: 0·013–4·50%) to divide patients into equal 'high' and 'low' *SLC22A1* expression groups (n = 47 and 46 respectively), no significant differences between these two groups were found by univariate analysis of achievement of *BCR-ABL1* RT-qPCR levels of 10%, 1%, 0·1% (on IS) or imatinib failure (P = 0·77, 0·45, 0·82 and 0·50, respectively). This was also the case for subgroup analysis of each of the two cohorts separately, when analysis was restricted to patients treated with 400 mg imatinib only and when thresholds other than the median were used to divide patients into high and low groups.

Exon 7 sequencing data

Sequencing of exon 7 of *SLC22A1* was carried out in 109 patients from the SPIRIT-1 cohort and 47 from the ICHNT cohort. Three SNPs and two insertions/deletions were identified (Fig 1 and Table II). Heterozygosity for 1201G>A (401G>S, rs34130495) was only found in nine patients and 1239G>A (a synonymous SNP, rs34888879) was only seen in one, with no patients homozygous for either SNP. 1222G>A (408V>M, rs628031) was a commonly observed variant, with an allelic frequency of 40%. The intronic 8 base pair (bp) insertion of TGGTAAGT (rs113569197, a duplication of the preceding 8 bp) at the 3′ end of exon 7 and the 3 nucleotide deletion (ATG), which leads to the loss of a methionine at position 420, (M420del, rs35191146) were mutually exclusive in a given allele for all patients tested.

For the purposes of further discussion, we will refer to alleles with the 8 bp insertion as the 8^+ allele (allelic frequency: 40%) and those with 3 bp deletion as the 3^- allele (allelic frequency: 18%). 8^+ was homozygous in all cases of

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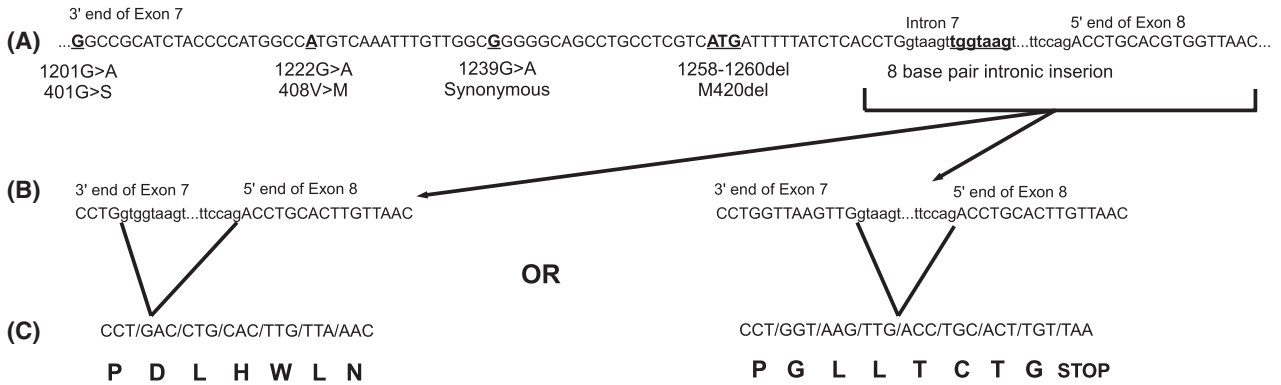

Fig 1. Polymorphisms in *SLC22A1* exon 7. (A) Polymorphism locations, nomenclature and amino acid changes. B and C show alternate splicing at the marked site in A. (B) Alternative splice donor sites at the 3′ end of exon 7. (C) cDNA and protein sequences resulting from alternative splicing.

Table II. Summary of polymorphisms identified in *SLC22A1* exon 7.

		Frequency		
Polymorphism	Other nomenclature	Wildtype	Heterozygous	Variant
1201G>A	401G>S, rs34130495	147 (94%) GG	9 (6%) AG	0 (0%) AA
1222G>A	408V>M, rs628031	54 (35%) GG	80 (51%) AG	22 (14%) AA
1239G>A	rs34888879	155 (99·4%) GG	1 (0·6%) AG	0 (0%) AA
1258-1260del (ATG) (3^-)	M420del, rs35191146	104 (67%) (ATG/ATG)	49 (31%) (ATG/3^-)	3 (2%) ($3^-/3^-$)
TGGTAAGT insertion (8^+)	rs113569197	54 (35%) ($8^-/8^-$)	80 (51%) ($8^+/8^-$)	22 (14%) ($8^+/8^+$)

8^+: insertion of 8 base pairs (bp). 3^-: 3 bp deletion. N: no 8 bp insertion or 3 bp deletion. 8^-: no 8 bp insertion (i.e. either N or 3^-).

1222A/A (M408), but was never seen in cases of 1222G/G (V408) and was found in a heterozygous configuration in all 1222A/G heterozygotes, suggesting that 1222A (M408) and the 8 nucleotide insertion (8^+) are consistently co-inherited in *cis*. These observations were replicated in four cell lines (K562, KCL22, BV173 and LAMA84), samples from a cohort of 20 newly diagnosed patients from Adelaide, Australia, and in 354 patients enrolled in the SPIRIT-2 trial. Patients can therefore be classified into six possible genotypes 8^+8^+, 8^+N, 8^+3^-, NN, N3^- and 3^-3^- (where N represents an allele with neither 8^+ or 3^-; allelic frequency 42%).

cDNA amplicon length analysis and sequencing

For the cDNA primers designed to amplify exon 7 through to the start of exon 8 of *SLC22A1* (forward CGG ACT CTG TGC TCT ATC AGG and reverse GTA CAG CTC AGC ATT CAC CA), the expected amplicon length for the N allele is 303 base pairs and that of the 3^- allele is 300 base pairs. Given the intronic nature of the 8 nucleotide insertion (8^+), transcript lengths were not predicted to differ between the N or 8^+ alleles. However, when transcript lengths were assessed using cDNA from 50 patients using high performance capillary electrophoresis on fluorescently labelled cDNA, a transcript length of 311 bp was found, in addition to those of 303 bp, in patients with 8^+ alleles (Table III). This indicated the presence of a longer mRNA transcript, which was confirmed by Sanger sequencing (in 12 patients) to be due to the presence of an additional 8 bp sequence (GTAAGTTG) at the 3′end of exon 7. This was consistent with the use of a duplicated 'gt' donor site downstream of the original donor site. This leads to a frame shift and the introduction of a premature stop codon after codon 429 (Fig 1B, C).

Clinical significance of exon 7 SNPs and deletions

It is not possible to draw any definite conclusions about patients with AG at 1201, due to the small number of patients with this SNP. As patients with 8^+8^+ and 8^+N were found to produce the same cDNA transcript profiles (i.e. the normal 303 bp transcript and the 311 bp transcript with a premature stop codon), these patients were analysed together (Group 1, $n = 76$) and compared to the remaining patients (8^+3^-, NN, N3^- and 3^-3^-; Group 2, $n = 80$). Significant differences were observed between these two groups for the probabilities of achieving RT-qPCR levels of 1% and 0·1% over time, but not for the rates of imatinib failure (Table III and Fig 2A–C).

For the SPIRIT-1 cohort alone (but not the ICHNT cohort), there were significant differences in achievement of RT-qPCR levels of 10%, 1% and 0·1% and imatinib failure rates for the two genotype groups defined above ($P = 0·01$, 0·002, 0·0001 and 0·02, respectively; data not shown) with achievement of 1% and 0·1% in Groups 1 ($n = 52$) and

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Table III. Summary of the six observed 1222G>A and 1260del genotypes, cDNA transcripts and clinical response.

Genotype	Frequency (percentage of total)	cDNA transcript lengths (bp)			Probability of 1% by 1 year (95% confidence interval), (%)	Probability of 0·1% by 1 year (95% confidence interval), (%)	Probability of imatinib failure by 5 years (95% confidence interval), (%)
$8^+ 8^+$	22/156 (14)		303	311	68 (41–83)	32 (10–49)	36 (13–54)
8^+N	54/156 (35)		303	311	67 (52–77)	28 (15–39)	31 (18–43)
8^+3^-	26/156 (17)	300	303	311	77 (53–89)	35 (23–51)	27 (8–42)
NN	28/156 (18)		303		82 (60–92)	46 (24–52)	29 (7–46)
N3^-	23/156 (15)	300	303		70 (44–84)	52 (27–69)	22 (3–38)
3^-3^-	3/156 (2)	300			100	67 (0–93)	33 (0–70)
$8^+8^+/8^+$N	76/156 (49)		303	311	67 (55–76)	29 (18–39)	33 (21–43)
8^+3^-/N3^-/3^-3^-/NN	80/156 (51)	Multiple			77 (66–85)	44 (32–54)	27 (16–37)

2 (n = 57) at 1 year of 63% vs. 84% (P = 0·02) and 25% vs. 51% (P = 0·006), respectively. Analysis of the SPIRIT-1 group also demonstrated superior outcomes for the N/N group in particular (n = 20), when compared to the 3^-3^-/N3^-/3^-8^+ (n = 35) and $8^+8^+/8^+$N (n = 52) groups (Fig 2D–F), suggesting a deleterious role of both the 8 bp insertion and the 3 bp deletion. Multivariate analysis was carried out using Cox proportional hazards regression modelling using the variables: *SLC22A1* SNP Group 1 or 2 (as defined above), *SLC22A1* expression (high or low), sex, age, imatinib dose (400 or 800 mg) and cohort. For the total cohort (n = 148), SNP group was found to be an independent predictor for achievement of RT-qPCR levels of 1% and 0·1% at 12 months (P = 0·018, 0·017 respectively), but not for imatinib failure (P = 0·59). These findings were not confirmed when the ICHNT cohort was assessed alone (n = 44), and this was the case whether or not the Sokal score was included in the model. Finally, for the SPIRIT-1 cohort alone (for which data on Sokal score were not available, n = 101), the SNP group was found to be predictive for RT-qPCR levels of 10%, 1%, 0·1% responses and imatinib failure (P = 0·010, 0·00014, 0·00046, 0·035, respectively). *SLC22A1* expression by RT-qPCR was not found to be an independent predictive factor in any of these analyses.

Discussion and conclusions

The human organic cation transporters (SLC22A1, 2 and 3) are a family of transmembrane proteins that are involved in the transport of endogenous cations and xenobiotics, such as metformin and acyclovir (Klaassen & Aleksunes, 2010). *SLC22A1* has seven known mRNA transcripts, only four of which are protein coding. The predominant splice isoform, *SLC22A1*-001 (OCT1a) has eleven exons and results in a functional 554 amino acid protein, while the other main isoform, of *SLC22A1* (OCT1b), does not contain exon 9 and leads to a non-functional 506 amino acid protein (Hayer *et al*, 1999; Thomas *et al*, 2004; http://www.ensembl.org/Homo_sapiens/Gene?g=ENSG00000175003). Thomas *et al*, (2004) demonstrated that imatinib was taken up into cells from a leukaemia cell line as an active process, dependent on SLC22A1. This has been demonstrated in other cell lines (Wang *et al*, 2008) and in primary patient specimens (White, 2006), but has not been replicated in all experimental systems (Hu *et al*, 2008).

Crossman *et al* (2005) measured *SLC22A1* mRNA levels in bone marrow mononuclear cells in 15 responders and 15 non-responders, defined by achievement of <65% Ph+ cells (major cytogenetic response) by 10 months, and found significantly higher *SLC22A1* levels in the group of responders. Wang *et al* (2008) measured *SLC22A1* mRNA by RT-qPCR in peripheral blood leucocytes of 70 imatinib-naive patients. They found that *SLC22A1* expression level (with the median expression as cut-off) predicted for progression-free survival (PFS) and achievement of CCyR at 6 months. It is worth noting that 20 out of 70 patients were not in chronic phase, and that 54 patients had received other lines of treatment prior to starting imatinib. Marin *et al* (2010) similarly classified patients into two groups according to expression of *SLC22A1* by RT-qPCR. This was measured at diagnosis in a cohort of 99 patients who achieved CCyR and who remained on imatinib for a number of years. Higher *SLC22A1* levels at diagnosis were found to predict for 3 and 4·5 log degrees of molecular remission (MR3 and MR4·5, respectively). Similarly, Nardinelli *et al* (2012) found higher rates of MR3 at 12 and 18 months and MR4·5 at any time point for patients with expression of *SLC22A1* (by RT-qPCR) greater than the median, while Zhong *et al* (2012) found higher expression of *SLC22A1* by RT-qPCR in the bone marrow of those who achieved cytogenetic responses. Razga *et al* (2011), however, performed a similar analysis in 30 patients using pre-treatment bone marrow and peripheral blood specimens and found no correlation between *SLC22A1* expression group and response at 6 or 12 months (according to ELN criteria). White *et al* (2007, 2010a) were also unable to correlate mRNA levels with the half maximal inhibitory concentration of imatinib (IC_{50}(IM)) or outcome.

Given that the level of activity of SLC22A1 may be due not only to differences in *SLC22A1* expression but also to

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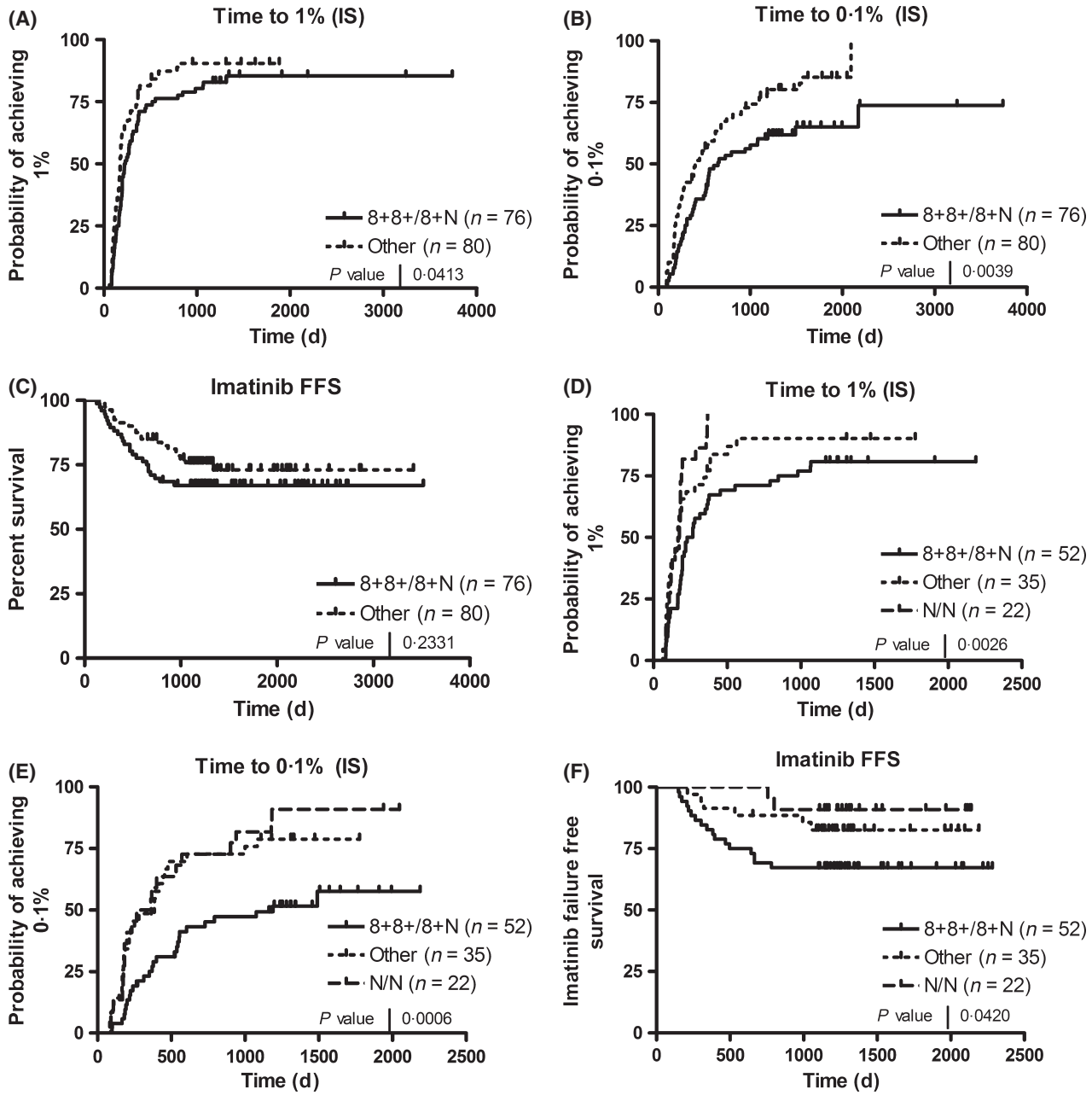

Fig 2. Comparisons of times to 1% (A, D), 0·1% (B, E) and imatinib failure-free survival (FFS) (C, F) according to genotype. A–C show results for the total cohort, and D–F for the SPIRIT-1 cohort alone. IS, International scale.

differences in translation, amino acid sequence or post-translational modification, White *et al* (2007) assessed the functional activity of *SLC22A1* in 56 patients from the Australian TIDEL (Therapeutic Intensification in *DE-novo* Leukemia) study by measuring imatinib intracellular uptake and retention (IUR) in peripheral blood mononuclear cells treated or not with the SLC22A1 inhibitor prazocin. 'SLC22A1 activity' was found to correlate with IC_{50} of imatinib and achievement of MR3 by 24 months. A subsequent study assessed these patients at a later time-point (60 months), demonstrating that the difference in the achievement of MR3 persisted and that poorer rates of MR4·5, event-free survival and overall survival were seen in the patients with low SLC22A1 activity (White *et al*, 2010a). These findings were replicated in samples from the TOPS study in which patients were randomized to either 400 or 800 mg daily of imatinib (White *et al*, 2012).

There are a number of known non-synonymous SNPs in the *SLC22A1* gene, several of which have been shown to affect protein function (Ahlin *et al*, 2010). Several studies have reported an association between *SLC22A1* SNPs and clinical outcome. Kim *et al* (2009) used Sequenom MassArray

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techniques to determine that the GG genotype for the C480G SNP in exon 2 (rs683369) correlated with lack or loss of response, while Bazeos *et al* (2010) reported that the presence of 1201G>A in exon 7 (rs34130495) was associated with improved rates of MR3. Takahashi *et al* (2010) found a correlation between possession of the GG genotype at 1222 (rs628031), also in exon 7, and achievement of MMR. Giannoudis *et al* (2012) recently reported improved outcome in patients with 1222G compared to 1222A, but only in the context of the presence of a 3 bp deletion at position 1260 (420del in exon 7), which, they speculated, causes reduced SLC22A1 activity and correlates with poorer outcome.

The present study sought to minimize the confounding factors as far as possible by selecting a cohort of patients who were previously untreated, with a median follow-up of 4 years, by using only diagnostic pre-treatment specimens, using multiple reference genes which were internally validated and by using primer and probe sequences that did not bind areas that contained known SNPs. No significant differences were found between 'high' and 'low' expressing groups and this was found to be the case even on subgroup analysis and when alternative thresholds were used. Exon 7 of the *SLC22A1* gene was sequenced to look for SNPs because, as noted above, three polymorphisms on this exon have been implicated in changes in SLC22A1 protein function and patient outcome. We also sought to explore whether variations in the SNPs at this location in patients from previous studies could be another explanation for the observed discrepancies.

An association was found between 1222G>A and an intronic 8 bp insertion (subsequently documented by Singh *et al* (2012) and Tarasova *et al* (2012) and replicated by our group in a cohort of over 300 patients from the SPIRIT-2 cohort), and this lead to an additional RNA variant with a premature stop codon. This is a transcript that has not been previously described and results from alternate splicing at the duplicated donor site. Patients homozygous for this insertion ($8^+/8^+$) produce the same two transcripts as heterozygotes (8^+N) and, collectively, these patients were found to have worse responses to imatinib compared to the remaining genotypes (N/N, N/3^-, $3^-/8^+$ and $3^-/3^-$). The best responses were seen in with the N allele (1222GG; lacking both the 8 bp insertion and 3 bp deletion, M420del). This is in keeping with Singh *et al* (2012), who correlated a haplotype containing 1222G>A and two intronic SNPs with lower imatinib trough levels, Takahashi *et al* (2010), who showed improved rates of MMR for 1222GG patients and with Giannoudis *et al* (2012), who showed inferior responses in the presence of 420del, which could be ameliorated by the presence of 1222G rather than 1222A allele. However, our results are not consistent with the assumption of this latter group that these two abnormalities can occur in *cis*. In our study, the 1222A allele was only seen in the presence of the 8 bp insertion, which was mutually exclusive for M420del.

Marin *et al* (2010) used primers for RT-qPCR that would be unaffected by these SNPs, but their probe would not be expected to bind sequences on the 3^- allele (because the sequence includes the ATG) or the alternate transcript produced with 8^+ (because the probe sequence assumes the normal exon-intron junction), and therefore the measured expression would be higher in N/N patients compared to patients with 3^- or 8^+. If possession of the N/N genotype offers superior outcomes, this may explain Marin's findings that increased expression correlates with improved imatinib response rates. Similarly, Wang *et al* (2008) used a forward primer at the 3′ end of exon 7 and the reverse primer on exon 9, which has the advantage of amplifying only the functional splice isoform of *SLC22A1*, but the forward primer also includes the ATG deleted in 3^- alleles. Again, this opens the possibility that correlations seen between outcome and expression are actually due to the presence of the polymorphism, as demonstrated when the forward primer is modified to avoid binding to that critical sequence (Giannoudis *et al*, 2012).

Collectively, our findings support the role of exon 7 polymorphisms in determining SLC22A1 protein levels/function and response to imatinib. It remains to be seen whether there is a clear correlation between genotype and imatinib uptake by leukaemic cells, and whether there are further mechanisms (whether host- or tumour-dependent) that determine the relative levels of different transcripts/isoforms (normal exon 7 splicing vs. splicing at the duplicate splice site; *SLC22A1*-001 *versus* *SLC22A1*-006) that might have functional relevance.

The 3 and 8 bp insertions/deletions can easily be detected using capillary electrophoresis or sequencing using either genomic or cDNA. Further prospective studies are required to assess whether information about an individual's genotype can be used to optimize the dosing or choice of TKI. Similarly, these findings may also be of relevance for patients treated with other drugs transported by SLC22A1, such as metformin in type II diabetes.

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Disclosure of conflicts of interest

The authors of this manuscript have no conflicts of interest to disclose.

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